

Wi-Fi Connectivity, Ping & Traceroute

Assignment Answers — Practical Network Testing

Important note about the practical results

The commands in this assignment must normally be run on the student's own laptop and phone. Because this PDF is prepared without access to those devices, the command outputs below are **sample/example outputs**. Replace the example IP addresses, hop count, and device names with the actual values from your network if required by your instructor.

1. Test communication between the laptop and phone using ping

Step 1: Connect both the Windows laptop and the friend's phone to the same Wi-Fi network.

Step 2: Find the phone's local IP address in the phone's Wi-Fi network details. For example, suppose the phone has IP address **192.168.1.105**.

Step 3: On the Windows laptop, open Command Prompt and run:

```
ping 192.168.1.105
```

Example result: Replies such as "Reply from 192.168.1.105: bytes=32 time=5ms TTL=64" show that the laptop can reach the phone over the local network. Four successful replies normally indicate basic IP connectivity.

2. Run traceroute to www.youtube.com

On Windows, open Command Prompt and run the following command:

```
tracert www.youtube.com
```

Example observation: The command may display several numbered hops. For example, if the last numbered response is hop 12, the route reached the destination in **12 reported hops**. The exact number can change because Internet routing changes over time, and some routers may not respond to traceroute probes.

Recorded example: 12 hops. This is an illustrative value, not a measurement from the student's device.

3. Disconnect one device and ping it again

Disconnect the phone from the Wi-Fi network and leave the laptop connected. Then run the same ping command:

```
ping 192.168.1.105
```

Typical Windows result: "Request timed out." If the destination is no longer reachable, the laptop may receive no reply from the phone. In some network configurations, a different message such as "Destination host unreachable" may appear.

Conclusion: The change from successful replies to timeouts/unreachable responses shows that the phone is no longer reachable at that local IP address while disconnected from the Wi-Fi.

4. List devices using arp -a

After the devices have communicated on the local network, open Command Prompt on Windows and run:

```
arp -a
```

IP address	MAC address	Type	Device / person (example)
192.168.1.1	A4-5E-60-12-34-01	dynamic	Home Wi-Fi router
192.168.1.102	B8-27-EB-45-67-02	dynamic	Student's laptop

IP address	MAC address	Type	Device / person (example)
192.168.1.105	7C-9E-BD-78-90-03	dynamic	Friend's Android phone
192.168.1.110	D0-37-45-23-45-04	dynamic	Smart TV

The table above is an example. The actual entries returned by **arp -a** depend on the devices that your laptop has recently communicated with. Match an IP/MAC address with a device by checking the device's Wi-Fi details or the router's connected-device list.

5. One-page network testing report

Objective: The purpose of the practical was to connect a laptop and a friend's phone to the same home Wi-Fi network and test local connectivity using ping and Internet routing using traceroute.

Network setup: First, the laptop and phone were connected to the same Wi-Fi network. The phone's local IP address was checked in its Wi-Fi settings. The laptop was then used as the main testing device because Windows Command Prompt provides the ping, tracert, and arp -a commands.

Ping test: The laptop was used to ping the phone's local IP address. In the example test, the command **ping 192.168.1.105** returned replies, showing that the two devices could communicate over the local network. The response time was only a few milliseconds, which is typical for a nearby device on a home Wi-Fi network.

Traceroute test: The command **tracert www.youtube.com** was then used to observe the path toward YouTube. The example output showed 12 reported hops. The number of hops is not fixed because Internet routing can change, and some routers may not answer traceroute requests.

Disconnected-device test: The phone was disconnected from Wi-Fi and the same ping command was run again. Instead of successful replies, a typical result is **Request timed out** or, depending on the network, **Destination host unreachable**. This demonstrates that the laptop could no longer reach the phone through the Wi-Fi network.

ARP check: Finally, **arp -a** was used to display recently learned local IP-to-MAC address mappings. These entries can help identify devices on the local network, although ARP does not necessarily show every device currently connected to the router.

Overall result: The tests demonstrate three useful network troubleshooting techniques. Ping checks basic reachability, tracert shows the sequence of network hops toward an Internet destination, and arp -a provides local IP and MAC address information. Any real command output should be recorded or captured as a screenshot and attached to the assignment.

Useful command summary

Purpose	Windows command	What it shows
Test local reachability	ping <IP address>	Whether the target responds and approximate latency
Trace Internet route	tracert www.youtube.com	Reported hops between the device and destination
View ARP entries	arp -a	Local IP-to-MAC address mappings known by the laptop

Submission tip: Replace the illustrative IP addresses and hop count with your actual command outputs, and insert screenshots of Command Prompt results where requested.